



$$\begin{aligned}
 & \sum_a \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) b(s) \\
 &= b(s) \nabla_\theta \sum_a \pi_\theta(a|s) = b(s) \nabla_\theta 1 = 0
 \end{aligned}$$